

The Algebraic Structure of Reality

Fundamental Fractal Geometric Field Theory (FFGFT)

*From $T^4/\mathbb{Z}_3$ to the Particles
of the Standard Model*

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<https://github.com/jpascher/T0-Time-Mass-Duality>

September 2026

Bibliographic Information

Johann Pascher: *The Algebraic Structure of Reality. FFGFT*.
First edition, September 2026.

Zenodo: DOI 10.5281/zenodo.17390357

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Part I

The Question

Chapter A

Do We Live in a Matrix?

The question is old. Plato posed it with the allegory of the cave. Descartes as radical doubt. And in 2003 Nick Bostrom gave it a modern form: if advanced civilizations can run simulations of their ancestors and choose to do so, then we almost certainly live in such a simulation.

For all readers

Imagine playing a video game so perfect that you cannot tell from inside whether it is a game. The physics would be consistent, the mathematics would work — you would have no way to distinguish. That is Bostrom's argument.

It is a serious question. And it is wrongly framed. "Simulation" is the wrong category — it presupposes a simulator, an external instance.

What if the universe does not *is* a simulation, but *has* a mathematical structure — deep, compelling, without external instance? What if the question is not: *who computes us*, but: *what is the structure the nature cannot avoid*?

What this book presents is such a structure: a compact four-dimensional torus T^4 with a threefold symmetry $\mathbb{Z}_3$, from which algebraically follows what must follow. No free parameters. No postulate about particles.

Chapter B

Why Not Ten?

The Standard Model has nineteen free parameters. It accepts them. Among them stands out one fact: three generations of fermions, nine fundamental families.

For all readers

Imagine finding a watch. It runs precisely — but you do not know why it is built the way it is. Why twelve hours, why these gears? The Standard Model is this watch. FFGFT asks: by what principle was it built?

The LEP experiment found no fourth light neutrino generation — experimentally. But “not found” is not “structurally impossible.”

FFGFT gives an algebraic answer: the space $T^4/\mathbb{Z}_3$ has exactly nine fixed points. Not eight. Not ten. Nine.

[B] Doc. 282

A tenth family is algebraically forbidden.

For all readers

When you draw a triangle, it has exactly three corners — not because you stopped, but because the structure determines it so. The $\mathbb{Z}_3$ circulant has three eigenvalues. A fourth generation would not add a fourth corner — it would destroy the structure.

A theory that says “there are nine” is impressive. A theory that says “there cannot be others” is a different class of statement.

Part II

The Geometry

Chapter C

What Mathematics Owes Nature

Three philosophical positions: Instrumentalism (mathematics as tool), Structuralism (Tegmark: the universe *is* mathematics), and Geometric Constitutionalism: a geometric ground structure forces algebraic consequences that match observation — without ontological claims.

FPGFT takes the third position. Its status system distinguishes:

[AXIOM] Declared, not derived.

[B] Algebraically proven, verified by test script.

[K] Derived from ξ , numerically checked.

[S] Sketched — plausible, derivation open.

[X] Excluded by proof or test script.

Chapter D

The Torus and $\tilde{T} \cdot m = 1$

A torus arises through identification: right edge = left edge, top = bottom. Locally flat, globally closed.

For all readers

An old video game where the figure disappears on the right and reappears on the left. Locally normal — globally connected. That is $T^4/\mathbb{Z}_3$.

From the compactification of one of the four circles follows the fundamental relation:

$$\tilde{T} \cdot m = 1$$

[K] Doc. 077

Mass is the inverse time scale of a mode. Heavy = short intrinsic time. In natural units: $E = mc^2 = E = m$ — the factor c^2 is a unit conversion factor, not a law of nature.

Chapter E

The Threefold Division: $\mathbb{Z}_3$ and the Nine Fixed Points

$\mathbb{Z}_3$ has three elements: identity, rotation by 120° , by 240° . Acting on T^4 it produces **nine fixed points**.

[B] Doc. 282

The hermitian $\mathbb{Z}_3$ circulant on $\mathbb{C}^3$ has exactly three eigenvalues $\lambda_k = a + b\omega^k + c\omega^{2k}$ ($\omega = e^{2\pi i/3}$) — three generations, no more.

The Frobenius split in $GF(27)^*$:

[B] Doc. 339

- Fixed points $\{+1, -1\}$ = massive sector (leptons, quarks)
- Eight three-cycles = eight gluons
- $\{f_1, f_2\}$ = Majorana (ν_1, ν_2); $\{f_3, f_4\}$ = Dirac
- ν_3 is Dirac — **falsifiable prediction**

[B] Doc. 343

Part III

The Algebra

Chapter F

Finite Fields

$GF(q)$: finite field with $q = p^n$ elements. Physical hierarchy:

Field	Order	Physical role
$GF(3)$	3	Color charge
$GF(9)^*$	8	Eight gluons
$GF(27)^*$	26	Complete fermion structure

The Legendre symbol $(k/13) \in \{+1, -1\}$ forces charge quantization $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$.

[B] Doc. 346

Chapter G

The Galois Lattice of Particle Masses

Mass formula: $m_i = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frac}}$
Winding numbers of leptons:

[K] Doc. 006

Particle	n_θ	n_ϕ	r_i	p_i
e	3	2	$4/3$	$3/2$
μ	5	4	$16/5$	1
τ	9	5	$25/9$	$2/3$

Key identity:

[K] Doc. 338

$$\frac{(r_\mu/r_e)^2}{\xi} = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 64 \cdot 25 \cdot 27 = 43200$$

Hence $\xi = 4/30000$ — algebraically forced, no free parameter.

Chapter H

Three Generations: Prohibition of the Fourth

The $\mathbb{Z}_3$ circulant is 3×3 — three eigenvalues, not four. Structural prohibition of the fourth generation.

[B] Doc. 282

Charge quantization and color from two Galois properties:

[B] Doc. 346

QR mod 13	$N \bmod 3$	Particle type
QR (neutral)	0 (singlet)	Neutrino
NQR (charged)	0 (singlet)	Charged lepton
NQR (charged)	1,2 (triplet)	Quark
QR + triplet	—	forbidden

Chapter I

Gluons, Color, and Confinement

The strong nuclear force is mediated by eight gluons. Why eight? The Standard Model sets this as a property of the gauge group $SU(3)$. FPGFT derives it.

Gluons as Galois Objects

In $GF(27)^*$ the Frobenius $\phi : x \mapsto x^3$ splits the 26 elements into fixed points $\{+1, -1\}$ (massive sector) and three-cycles. The four $\mathbb{Z}_{13}$ orbits combined with $\mathbb{Z}_2$ parity give:

[B] Doc. 339

$$4 \text{ orbits} \times 2 \text{ parities} = \mathbf{8} \text{ gluons}$$

The three elements of each orbit are the three colors $\{R, G, B\}$. The fixed field $GF(3)^* = \{1, 2\} \cong \mathbb{Z}_2$ yields the photon.

Structure element	Physical meaning
Fixed points $\{+1, -1\}$	Massive sector (fermions)
4 orbits $\times$ 2 parities	Eight gluons
3 elements per orbit	Three colors R, G, B
Fixed field $GF(3)^*$	Photon ($U(1)$ sector)

For all readers

Imagine 26 elements divided into eight triplets plus two fixed points. The eight groups are the gluons, the three positions the colors, the two fixed points the massive matter. All from a single group structure — no free parameter.

Confinement as $\mathbb{Z}_3$ Triality Selection

Why has no one ever seen a single quark? In FFGFT the answer is algebraic: only $\mathbb{Z}_3$ -**invariant** objects — with total triality $N \bmod 3 = 0$ — can exit the torus and be observed.

[B] Doc. 325

- Single quark: $N \bmod 3 = 1$ or 2 — **not observable**
- Three quarks (baryon): $1 + 1 + 1 \equiv 0$ — **observable**
- Quark+antiquark (meson): $1 + 2 \equiv 0$ — **observable**

For all readers

A doorman rule: only groups whose sum is a multiple of three may enter. A quark with ticket "1" cannot enter. Three quarks with "1", "1", "1" — sum 3 — can enter. That is $\mathbb{Z}_3$ triality. That is confinement.

The Proton

A proton (uud) carries colors $\{R, G, B\}$ — $\mathbb{Z}_3$ -invariant. The charge $+1 = 2 \times (+2/3) + 1 \times (-1/3)$ follows from charge quantization via Legendre symbol.

[B] Doc. 346

The proton mass 938 MeV arises mainly from the gluon field (lattice QCD), not from quark masses. The Galois derivation of the exact mass is open.

[S]

Chapter J

Antiparticles, Spin, and Chirality: States, Not Doublings

When counting Standard Model particles, one quickly encounters confusion. There are twelve fermions — but then antiparticles too: the positron for the electron, the antiquark for the quark. Does that make 24? And then spin-up and spin-down for each. Does that make 48?

The answer is: no. And the explanation lies in the algebra.

What an Antiparticle Is

An antiparticle is not a different particle. It is the same algebraic object — with reversed charge. The positron has the same mass as the electron, the same spin, the same interaction strength. Only the electric charge is opposite: $+1$ instead of -1 .

In $GF(27)^*$ this difference corresponds to the $\mathbb{Z}_2$ parity — the factor $\mathbb{Z}_2$ in the decomposition $\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$. Particle and antiparticle are the two $\mathbb{Z}_2$ elements of the same orbit.

[B] Doc. 339

They are not two separate objects. They are one algebraic object with two states — like a coin with two sides, not two different coins.

For all readers

Imagine a clock. The hands can turn clockwise or counterclockwise. That is not a second mechanism — it is the same mechanism in two possible states. Particle and antiparticle are the same algebraic object, once with charge $+q$, once with $-q$. The $\mathbb{Z}_2$ parity in $GF(27)^$ is exactly this reversal state.*

What Spin Is

Spin is even more fundamental. It is not a spinning ball — that historical picture failed in 1925 because it would require unphysical rotation speeds. Spin is an algebraic property following from the structure of space.

In the Clifford algebra $Cl(6)$ — the algebraic description of the six-dimensional internal space of FFGFT — every state has a grade r determining how it transforms under rotations. Spin-1/2 particles correspond to grade-1 objects (vectors) in Clifford space.

The key: a spin-1/2 particle requires a **rotation by 720°** to return to itself — not 360° . This is not a paradox. It follows from the algebraic structure:

$$\gamma_5 \cdot X = (-1)^r \cdot X$$

For odd grades r : $\gamma_5 X = -X$ — this is **left-handed**. For even grades: $+X$ — **right-handed**.

[B] Doc. 349

Spin-up and spin-down are also not two different particles. They are two projections of the same algebraic object onto different spatial directions.

For all readers

A shadow on a wall can be long or short, depending on how the light falls. The object is the same — the shadow is a projection. Spin-up and spin-down are projections of the same spinor onto different measurement directions. Not two objects — one object, two measurement states.

What Chirality Is

The weak nuclear force behaves strangely: it acts only on left-handed particles and right-handed antiparticles. This is one of nature's most peculiar asymmetries — **parity violation**, discovered in 1956.

In FFGFT this asymmetry follows from the $Cl(6)$ grade parity — from the same γ_5 operator that defines spin. The **Su sector** (even $Cl(6)$ grades) is right-handed. The **Sd sector** (odd $Cl(6)$ grades) is left-handed. And the weak force $SU(2)_L$ couples only to the Sd sector — algebraically forced, not postulated.

[B] Doc. 349

The right-handed neutrino ν_R is structurally absent: in the Sd sector there is no room for it. This is consistent with the prediction from the neutrinos chapter: ν_3 is Dirac, but ν_R as an independent degree of freedom does not exist in $GF(27)^*$.

[B]/[S]
Doc. 349

The Complete Count

What are the "12 particles" really?

Property	Algebraic origin	Number of states
Particle families	9 fixed points of $T^4/\mathbb{Z}_3$	9
Color	$N \bmod 3$ in $GF(3)$	$\times 3$ for quarks
Particle/antiparticle	$\mathbb{Z}_2$ parity in $GF(27)^*$	$\times 2$
Chirality	$Cl(6)$ grade parity	$\times 2$
Spin projection	Measurement direction	2 states

None of this is an independent doubling. Everything is states of the same algebraic object — different projections, parities, grades within a single structure.

The "12 fermions" of the Standard Model are in FFGFT: 3 leptons (QR sector, colorless) + 3×3 quarks (NQR sector, 3 colors) = 12 *family-color combinations*. Antiparticles, spin, and chirality are not additional objects — they are algebraic states within the same Galois structure.

Chapter K

The Higgs Boson: Particle or Vacuum Artifact?

In July 2012 CERN announced the discovery of the Higgs boson — a particle with mass 125 GeV, predicted by the Standard Model since 1964. A triumph. But a question was left unanswered in the celebration:

What is the Higgs boson, really?

What the Standard Model Says

In the Standard Model there is a Higgs field Φ permeating all of space. This field has a **vacuum expectation value** $\langle \Phi \rangle = v \approx 246$ GeV — it is non-zero in the vacuum. Particle masses arise from interaction with this background: $m = y \cdot v$, where y is the Yukawa coupling.

The Higgs boson H^0 is the quantum oscillation of this field around its equilibrium — the excitation observed at the LHC in 2012.

Why does the vacuum have this non-vanishing value? Because the Higgs potential $V(\Phi) = \mu^2 |\Phi|^2 + \lambda |\Phi|^4$ with $\mu^2 < 0$ has its minimum not at $\Phi = 0$ but at $|\Phi| = v$. Why is $\mu^2 < 0$? The Standard Model answers: *it is set that way. That is how it is.*

For all readers

Imagine a valley. The Standard Model says: the universe sits in the valley because the valley is there. Why is the valley there? Because the potential has this shape. Why does it have this shape? No answer. FFGFT asks differently: what geometry forces this valley?

What FFGFT Says

In FFGFT, $\tilde{T} \cdot m = 1$ is not a postulate about fields — it is a compactification relation on T^4 . And this relation has a direct consequence: **a vacuum with $m = 0$ is algebraically excluded.**

[B] Doc. 354

If $m = 0$, then $\tilde{T} \rightarrow \infty$ — an infinitely long intrinsic time scale. That is physically unrealizable on a compact torus. The torus *forces* mass in the vacuum. What the Standard Model sets through $\mu^2 < 0$ follows in FFGFT from geometry.

The Higgs field Φ and the time field $\tilde{T}$ are **inverse to each other:**

$$\tilde{T}(x, t) = \frac{1}{m(x, t)} = \frac{1}{y \cdot (v + h(x, t))}$$

[B] Doc. 354

The time field is the inverse Higgs field. The Higgs mechanism is the *flat projection* of the compact T^4/Z_3 topology into the language of quantum field theory.

The spontaneous symmetry breaking $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{EM}}$ follows from three Galois building blocks:

[B] Doc. 355

- $\tilde{T} \cdot m = 1$ forces $\langle \Phi \rangle \neq 0$
- Z_3 Galois structure forces $Q_{\text{vac}} = 0$ (neutral vacuum)
- $Q = 0$ selects $U(1)_{\text{EM}}$ as residual group

Does the Higgs Boson Exist?

This question has a precise answer in FFGFT — in two parts.

Part 1: The Higgs field exists. The non-vanishing vacuum expectation value is forced, not postulated. [B]

Part 2: Whether the Higgs boson H^0 is an independent excitation or an artifact of the flat description — that is open.

[S] Doc. 354

The particle observed at 125 GeV is real [X]. The question is what it *is*: A true excitation of a fundamental field? Or the representation

of a torus topology in the language of flat quantum field theory — useful as a description, but not fundamental?

For all readers

Sound is real — we hear it. But sound is not fundamental: it is a pressure wave in air. In a theory without air there is no sound. The Higgs boson may be similar: real as an observation, but not fundamental — a wave in a vacuum field that is itself a projection artifact. FFGFT says: possibly. Not yet proved.

The most honest statement FFGFT can make: the Higgs mechanism follows from geometry [B]. The Higgs boson as a particle is experimentally established [X]. Whether it is fundamental or a projection — open [S].

Chapter L

Neutrinos: Majorana or Dirac?

Majorana criterion: $k \mapsto k^{-1} \bmod 13$. Orbit 3 = {4, 10, 12} self-inverse: Majorana. Orbit 4 = {7, 8, 11} not self-inverse: Dirac.

[B] Doc. 340

Neutrino masses:

[K] Doc. 340

Neutrino	Mass	Galois origin
ν_1	4.54 meV	Orbit 1, m_ν
ν_2	9.81 meV	Orbit 3, $\sqrt{14/3} m_\nu$
ν_3	49.96 meV	Orbit 4, $11 m_\nu$

Δm^2_{atm} : dev. 1.0 %; Δm^2_{sol} : dev. 0.46 %. Prediction: $m_{ee} \leq 14.4 \text{ meV}$ (testable by KamLAND-Zen).

Part IV

The Constants

Chapter M

One Parameter, All Constants

Constant	FFGFT	PDG	Dev.
$1/\alpha$	$3700/27 = 137.037$	137.036	7.6 ppm
$\sin^2 \theta_W$	0.2305	0.23122	0.06 %
m_μ/m_e	207.846	206.768	0.52 %
Δm_{atm}^2	$2.476 \times 10^{-3} \text{ eV}^2$	$2.500 \times 10^{-3} \text{ eV}^2$	1.0 %
Δm_{sol}^2	$7.565 \times 10^{-5} \text{ eV}^2$	$7.530 \times 10^{-5} \text{ eV}^2$	0.46 %
Σm_ν	64.3 MeV	< 120 MeV	consistent
m_τ	1776.78 MeV	$1776.86 \pm 0.12 \text{ MeV}$	0.4σ

Chapter N

Leptons and Their Residuals

Residuals ε_i in units of ξ : $+39\xi$, $+78\xi$, $+117\xi$ — systematically falling. [K] Doc. 352
QED self-energy as cause: excluded at 112σ . [K] Doc. 352
Open bridge R113: physical attribution open. [S] R113

Chapter 0

The Zeta Function of the Torus

$$\zeta_{T^4/\mathbb{Z}_3}(s) = (1 - 3^{-s}) \cdot \zeta(s)$$

[B] Doc. 343

New zeros at $s = 2\pi i k / \ln 3$ on $\text{Re}(s) = 0$ — signature of $\mathbb{Z}_3$ symmetry. No resonance with Riemann zeros. FFGFT touches the Riemann Hypothesis — does not solve it.

Part V

Representation and Being

Chapter P

Representation, Not Ontology

The Galois classes $\{f_1, f_2, f_3, f_4\}$ *represent* particles — they are not particles. Geometry forces algebra. Algebra matches observation. Ontology remains open.

Chapter Q

Superposition Without Time

No time operator in QM (Pauli 1933). Superposition is timeless.

In FFGFT: $\hat{T}_i = 1/m_i$ mode-bound. Cross-mode superposition creates incompatible time scales. The measurement act discards [B] Doc. 334 the winding number — the arrow of time has a geometric reason. [K] Doc. 333

Chapter R

Geometry as Deepest Basis

Constitution type: **Geometry [AXIOM] → Algebra [B] → Empirics [K]**.

Lorentz invariance: locally preserved, globally topological.

[K] Doc. 312

Bell correlations as holonomy on T^4 .

[B] Doc. 230

Part VI

Review and Outlook

Chapter S

What Is Proved, Sketched, Open

Proved [B]

Nine fixed points. Three generations. Prohibition of fourth. Eight gluons. Majorana $\nu_{1,2}$. Dirac ν_3 . Charge quantization. Gell-Mann-Nishijima. Prohibition neutral+colored. Coupling matrix $f_3 \times f_4$. Entries $1/\sqrt{2}$ in $f_1 \times f_3$. All Yukawa coefficients in Galois lattice.

Open [S]

R113 bridge (falling residuals). τ_μ direct. Cosmic exponent P20. CMB peaks. Quark sector complete.

Chapter T

The Matrix Question Answered

Simulation Universe as artifact — no simulator in FFGFT.

Tegmark One of infinitely many structures — FFGFT makes concrete predictions.

Zuse/Wolfram Discrete rules — geometric consistency missing there.

FFGFT The universe is like a proof. It cannot be otherwise.

For all readers

Simulation — like a video game. Tegmark — like a number. Zuse — like an algorithm. FFGFT — like a proof. It cannot be otherwise.

Falsifiable predictions:

- ν_3 is Dirac (no neutrinoless double beta decay for ν_3)
- $\Sigma m_\nu \approx 64.3$ meV (KATRIN, Euclid)
- $m_\tau = 1776.78$ MeV (Belle II, BESIII)
- No fourth quark generation (LHC)

Feynman wrote 1/137 on the wall and asked: Why?

FFGFT writes: Because $3700/27$.

The universe is not a simulation. It is an algebraic structure — and this structure permits no other particles than those we know. That is less than ontology. And more than physics has had before.

Appendices

Appendix 1

Status System and Document Numbering

Status Markers

- [AXIOM]** Declared, not derived. In FFGFT: $T^4/\mathbb{Z}_3$.
- [B]** Algebraically proved, verified by test script.
- [K]** Derived from ξ , numerically checked against PDG.
- [S]** Sketched — plausible, derivation open.
- [X]** Excluded.
- [Q]** External primary source.

Markers appear as marginal notes (right margin). Correction register: Doc. 190.

Access

- GitHub: <https://github.com/jpascher/T0-Time-Mass-Duality>
- Zenodo: [DOI 10.5281/zenodo.17390357](https://doi.org/10.5281/zenodo.17390357)
- ORCID: [0009-0000-6518-4064](https://orcid.org/0009-0000-6518-4064)

Appendix 2

Mathematical Foundations

Galois Fields

$GF(p^n)$: finite field. $GF(3) = \{0, 1, 2\} \bmod 3$. $GF(9)$: extension with $i^2 = 2$.
 $GF(27)$: cubic extension.

Frobenius Automorphism

$\phi : x \mapsto x^3$ on $GF(27)$. Order 3. Fixed points $\{+1, -1\}$ = massive sector.

$\mathbb{Z}_3$ Circulant

$C = \begin{pmatrix} a & b & c \\ c & a & b \\ b & c & a \end{pmatrix}$, eigenvalues $\lambda_k = a + b\omega^k + c\omega^{2k}$, $\omega = e^{2\pi i/3}$.

$$\tilde{T} \cdot m = 1$$

Compactification relation: $R_m = \hbar/(mc^2)$.

ξ from Galois

$$\xi = \frac{(r_\mu/r_e)^2}{|GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)|} = \frac{144/25}{43200} = \frac{4}{30000}$$

Appendix 3

Key Documents

Doc.	Content
006	Yukawa ladder, winding numbers (r_i, p_i)
077	$E = mc^2 = E = m, c$ as convention
190	Correction register, open bridges
241	Two-layer architecture
282	$\mathbb{Z}_3$ circulant, mass operator
307/334	Time in QM, superposition
312	$\Lambda \rightarrow 1/R_{Hr}^2, 10^{123}$ decomposition
338	$(m_\mu/m_e)^2 = 43200$ [K]
339	Frobenius split massive/massless [B]
340	Δm^2 from $GF(27)^*$ [K]
342/343	Prime harmonics, orbifold zeta [B]
346–348	Charge, Gell-Mann-Nishijima, generations [B]
351/352	Lepton residuals, PDG model dependence [K]

Appendix 4

Open Bridges (September 2026)

1. **R113** [S]: Why do ε_i fall with generation?
2. τ_μ **direct** [S]: Convention-free test of weak sector.
3. **P20** [S]: Derive cosmic exponent 10 forward.
4. **CMB peaks** [S]: Sequence $\{1, 6, 14, 26\}$ from torus modes.
5. **Quark sector** [S]: Derive coefficients purely geometrically.
6. **CKM phase** [S]: $\delta_{CP} \approx 68^\circ$ from $GF(27)^*$.